

Section 2 Solutions

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4 Exercises

4.1 Preferences and Utility Functions

Suppose there are two goods in the world: apples (x) and oranges (y), and that our preferences can be represented by the following utility function: $u(x, y) = 2x + y$

1. If a consumer does not have any apples, how many oranges would he need to be indifferent to a consumption bundle containing 2 apples and 1 orange?

Answer: 5

2. Draw the indifference curves for when the utility equals 5 units and when the utility equals 7 units. Do these indifference curves cross? Can they ever cross? Why or why not?

Answer: The equations describing these curves are: $y = 7 - 2x$ and $y = 5 - 2x$ which are easy to draw. They can't cross, won't cross (different intercept, same slope). If they crossed, that would imply that a bundle of goods can have two different utilities, which makes no sense.

3. Prove that the preferences represented by this utility function are transitive. That is, show that if $(x_1, y_1) \succsim (x_2, y_2)$ and $(x_2, y_2) \succsim (x_3, y_3)$, then $(x_1, y_1) \succsim (x_3, y_3)$.

Answer:

$$2x_1 + y_1 \geq 2x_2 + y_2$$

$$2x_2 + y_2 \geq 2x_3 + y_3$$

by transitivity of $\geq$, it follows that:

$$2x_1 + y_1 \geq 2x_3 + y_3$$

*These notes are a consolidation of those of Jianmeng Lyu, Sam Leone, Pablo Munoz, Yassine Sbair Sassi, Jon Schellenberg, Nick Li, Katalin Springel, Jonas Tungodden, Dan Acland, Justin Gallagher, Mariana Carrera, Matt Leister, Ivan Balbuzanov, Anne Karing, and David Wu, Sam Wang, and Matteo Saccarola.

4.2 Simple IFT

The function $f(x; p) = 1 - xp - p \cdot \exp(x) = 0$, around $(x_0, p_0) = (0, 1)$. Use the implicit function theorem to find $\frac{dx}{dp}$.

Solution:

1. First, check that conditions are satisfied: (1) $f(x; p)$ is continuously differentiable around (x_0, p_0) .
(2) $f_x(x_0; p_0) = -p_0 - p_0 \exp(x_0) = -2 \neq 0$. Then,

$$\frac{dx^*}{dp} = -\frac{\frac{\partial f}{\partial p}}{\frac{\partial f}{\partial x}} = -\frac{-x - \exp(x)}{-p - p \cdot \exp(x)} = -\frac{x + \exp(x)}{p(1 + \exp(x))}$$

We can actually simplify this expression (although you don't have to). From $f(x; p) = 1 - xp - p \cdot \exp(x) = 0$ we know

$$x + \exp(x) = \frac{1}{p}, \quad 1 + \exp(x) = 1 + \frac{1 - xp}{p} = \frac{1 + (1 - x)p}{p}$$

Therefore,

$$\frac{dx^*}{dp} = -\frac{x + \exp(x)}{p(1 + \exp(x))} = -\frac{1}{p + (1 - x)p^2}$$

Note that you can only use IFT to derive $\frac{dx^*}{dp}$ because we can't write down the explicit function of $x^*(p)$.